

StockSense AI — Product Requirements

AI-Powered Inventory Forecasting & Replenishment Platform

Product Overview

StockSense AI is a decision-support platform for retailers and e-commerce businesses. It combines sales, inventory, supplier constraints, calendars and planning policies to generate demand forecasts and replenishment recommendations.

The product is designed around human approval. Recommendations are explainable, traceable and editable before material purchasing actions are taken.

Core Requirements

Data ingestion and quality monitoring; product and location master data; inventory visibility; configurable demand forecasting; forecast evaluation; replenishment calculation; safety stock; supplier lead-time constraints; exception management; purchase planning; approvals; scenario planning; dashboards; alerts; reporting; security and audit.

AI / ML Requirements

Forecast outputs carry model/version metadata and timestamps. Forecast quality is monitored with measures such as WAPE, MAE and bias. Model changes require benchmark evaluation and rollback capability.

New-product and sparse-history workflows can use configurable analogs or planner assumptions. Material forecast changes should surface explanatory signals rather than unsupported causal claims.

Business Workflow

Source data → validation → forecasting → projected inventory → replenishment recommendation → exception review → purchase plan → approval → export/integration → outcome measurement.

Success Measures

Forecast quality, stockout exposure, excess inventory, planning cycle time, recommendation acceptance, exception resolution time and data-quality failure rate. Guardrails include systematic model bias, bad recommendations and unauthorized access.

Prepared as a portfolio case-study artifact. Requirements and metrics are illustrative unless explicitly validated in a real implementation.